Linear Classification

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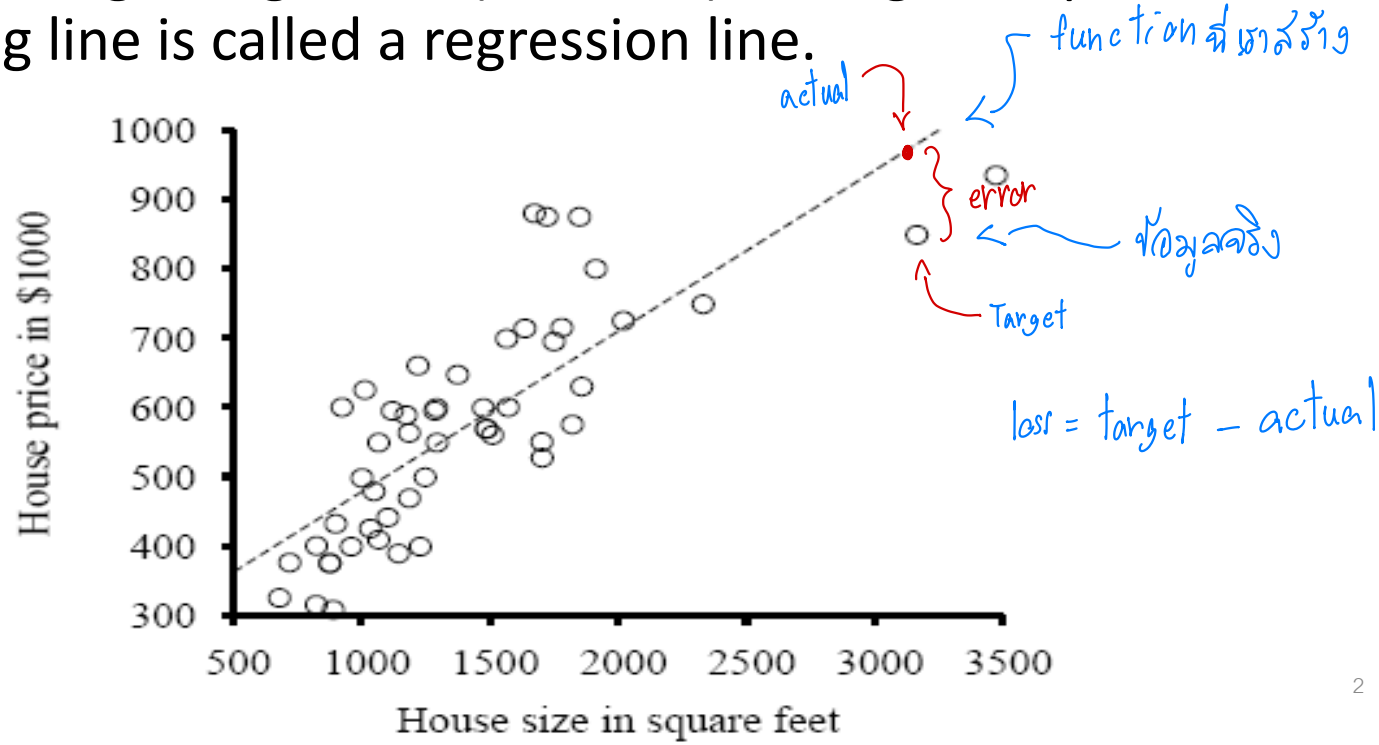
Regression output เป็น ต่อเลข ทศนิยม

classification output เป็น การแบ่งกลุ่มชัดเจน อาจจะมีได้มากกว่า 2

Regression and Classification with Linear Models

- Regression analysis is a statistical process for estimating the relationships between a dependent variable y and independent variable(s) x.
- Given data points of (x, y), linear regression consists of finding the best-fitting straight line (function) through the points. The best-fitting line is called a regression line.



1 variable

Univariate linear regression

y = ax+b

- Univariate linear function (a straight line) with input x and output y has the form $h_w(x) = w_1x+w_0$, where w_0 and w_1 are real-value coefficients (weights) to be learned.

  y ax+b

- To find h_w that best fit input data x, we have to find the values of the weights $[w_0,w_1]$ that minimize the loss (e.g. training errors).

ไม่รู้ว่าเป็นบวกหรือลบ

Target value

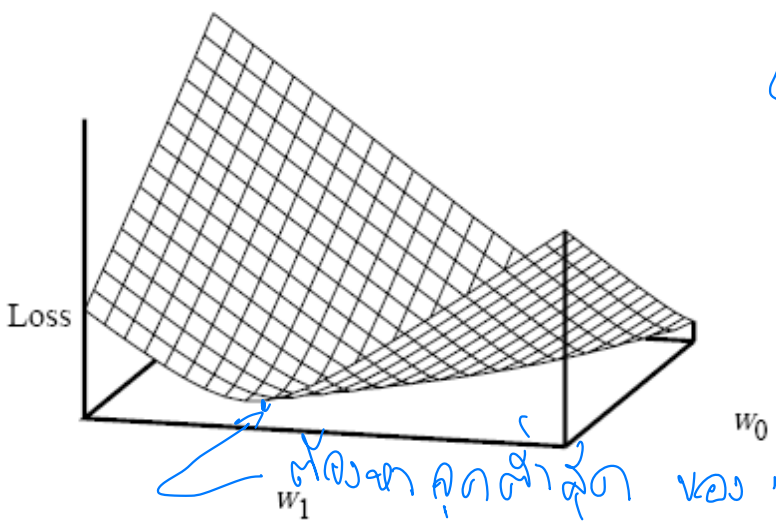
Number of examples

Actual value

- $\text{Loss}(h_w) = \sum_{j=1}^{N}\left(y_j - h_w(x_j)\right)^2 = \sum_{j=1}^{N}\left(y_j - (w_1x_j + w_0)\right)^2$

ใช้ abs ได้ แต่ กำลังสอง มี ข้อดี คือ

ถ้า error เยอะ ค่า loss จะเยอะแบบทวีคูณ
จึงบังคับให้ error อย่าเยอะ



ต้องหาจุดต่ำสุด ของ w_0 กับ w_1

Note that this loss function is convex, with a single global minimum.

instance ใน dataset (ทุก)

- The sum $\sum_{j=1}^{\boxed{N}}\left(y_j - (w_1 x_j + w_0)\right)^2$ is minimized when its partial derivatives with respect to w_0 and w_1 are zero.

$$\underbrace{\frac{\partial}{\partial w_0}\sum_{j=1}^{N}\left(y_j - (w_1 x_j + w_0)\right)^2 = 0 \ , \frac{\partial}{\partial w_1}\sum_{j=1}^{N}\left(y_j - (w_1 x_j + w_0)\right)^2 = 0}_{\text{Optimization problem: solve equations for w0 and w1}}$$

คือหาค่า loss

- For example in the above figure, the solution is w_1=0.232, w_0= 246, and the line with those weights is shown as a dashed line in the figure.

- In a general optimization problem, this optimization problem can be addressed by a **hill-climbing algorithm** that follows the gradient of the function to be optimized. We will use **gradient descent** to minimize the loss.

คอมชัน

ขยับเข้าใกล้

w = any point in the parameter space

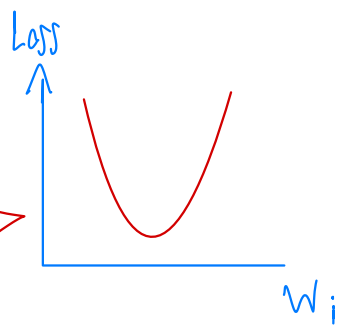
w = {w0, w1, w2, w3, …, wn}

loop until convergence → คือ w เปลี่ยนแปลงน้อย หรือ ไม่เปลี่ยนแปลง

for each w_i in w do

$$w_i \leftarrow w_i - \boxed{\propto} \boxed{\frac{\partial}{\partial w_i} Loss(\boldsymbol{w})} \quad (1)$$

learning rate



$Note$: ∝ is usually called the learning rate.

คอมชันเป็นบวก ต้องลด weight ลง

คอมชันเป็นลบ ต้องเพิ่ม weight

- For univariate regression, the loss function is a quadratic function, so the partial derivative will be a linear function.

$$\frac{\partial}{\partial w_i} Loss(\boldsymbol{w}) = \frac{\partial}{\partial w_i}(y - h_w(x))^2$$
$$= 2(y - h_w(x)) \times \frac{\partial}{\partial w_i}(y - h_w(x))$$
$$= 2(y - h_w(x)) \times \frac{\partial}{\partial w_i}(y - (w_1 x + w_0))$$

- Applying this to both w_0 and w_1 we get:

$$\frac{\partial}{\partial w_0} Loss(\boldsymbol{w}) = -2\big(y - h_w(x)\big)$$
$$\frac{\partial}{\partial w_1} Loss(\boldsymbol{w}) = -2\big(y - h_w(x)\big) \times x$$

$$w_i \leftarrow w_i - \propto \frac{\partial}{\partial w_i} Loss(\boldsymbol{w})$$

- Then, plugging this back into Equation (1), and folding the 2 into the unspecified learning rate ∝, we get the following learning rule for the weights:

$$w_0 = w_0 + \propto \left(y - h_w(x)\right)$$
$$w_1 = w_1 + \propto \left(y - h_w(x)\right) \times x$$

Target value Actual value

This is the weight updating for a single training example.

- For N training examples, we want to minimize the sum of the individual losses for every example (**batch gradient descent** learning rule).

$$w_0 = w_0 + \propto \sum_j \left(y_j - h_w(x_j)\right)$$
$$w_1 = w_1 + \propto \underbrace{\sum_j \left(y_j - h_w(x_j)\right) \times x_j}_{\text{Sum derivatives of the loss values from N training examples.}}$$

batch training

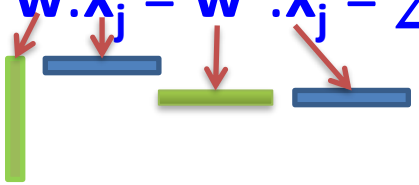
Multivariate linear regression

- We can easily extend to multivariate linear regression problems, in which each example x_j is an n-element vector. Our hypothesis space is:

$$h_{sw}(x_j) = w_0 + w_1\mathbf{x}_{j,1} + \ldots + w_n\mathbf{x}_{j,n} = w_0 + \sum_i w_i \boldsymbol{x}_{j,i}$$

- The w_0 term, *the intercept*, stands out as different from the others.
- If we introduce input attribute ($x_{j,0}$) which is defined as always equal to 1, then h is simply the dot product of the weights and the input vector:

$$h_{sw}(x_j) = \mathbf{w}.\mathbf{x}_j = \mathbf{w}^T.\mathbf{x}_j = \sum_i w_i x_{j,i}$$



- The update weight w_i by substitute $h_{sw}(x_j)$ into the equation in page 7:

Target value

Actual value

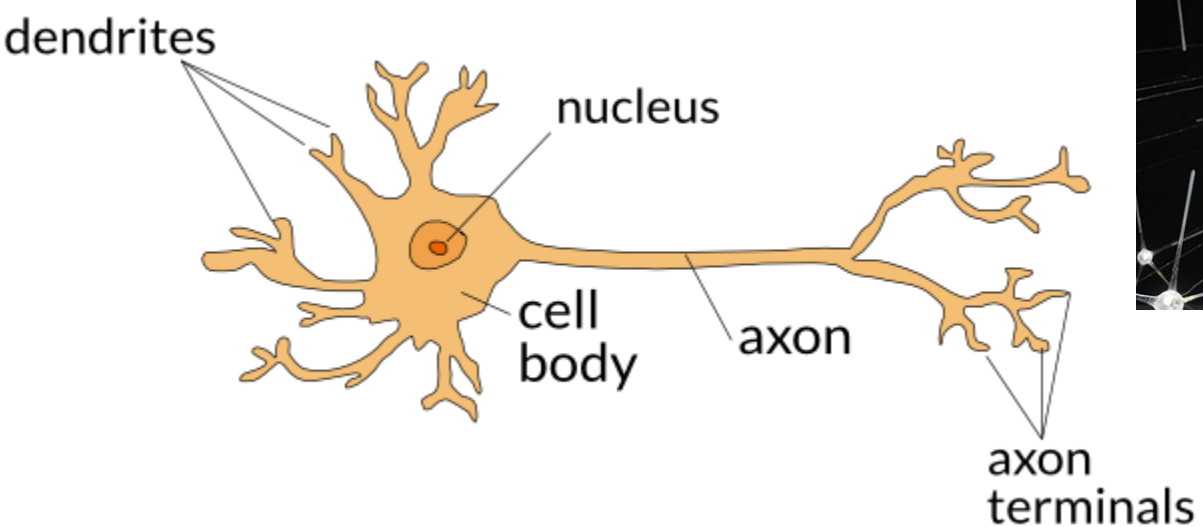
$$w_i = w_i + \propto \sum_j x_{j,i} \times \left(y_j - h_{sw}(x_j)\right)$$

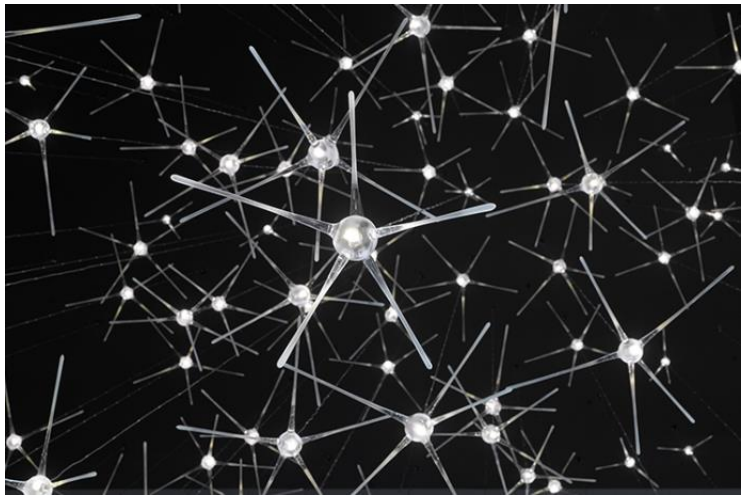
- This process is equivalent in finding the best vector of weights, **w***, that minimizes squared-error loss over the examples:

$$\boldsymbol{w}^* = \underset{w}{\operatorname{argmin}} \underbrace{\sum_j (y_j - h_{sw}(x_j))^2}_{\text{loss}}$$

Linear Classification With Perceptron

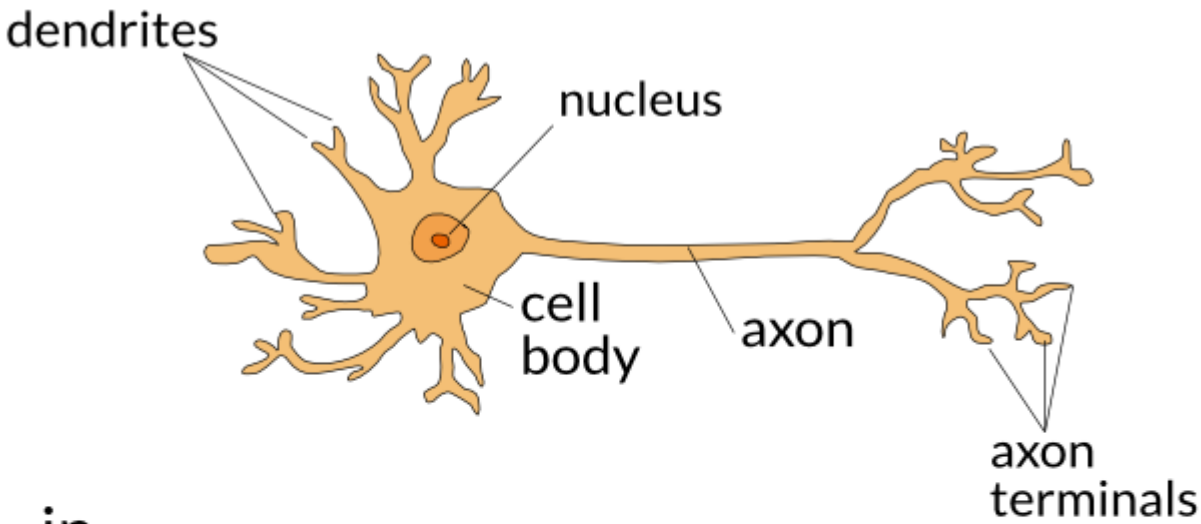
Biological neuron

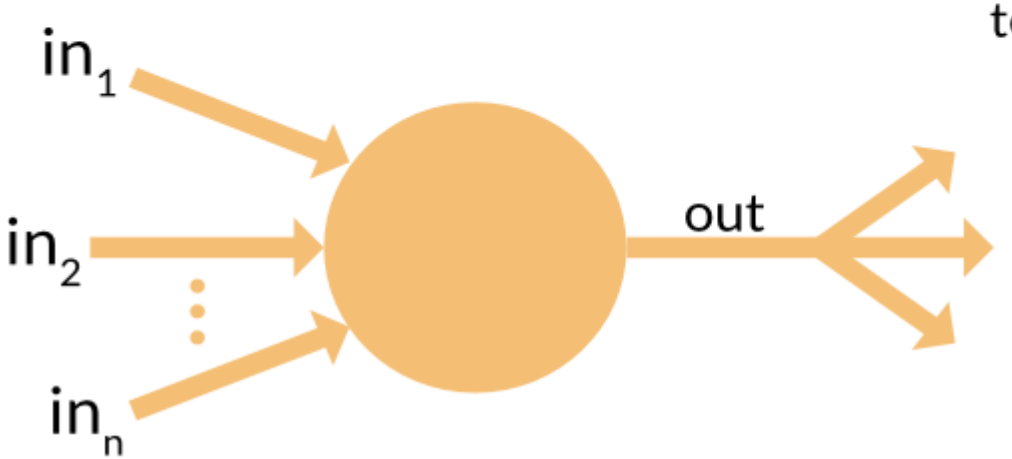




- Dendrites receive signals
- Axon sends signals out to other neurons

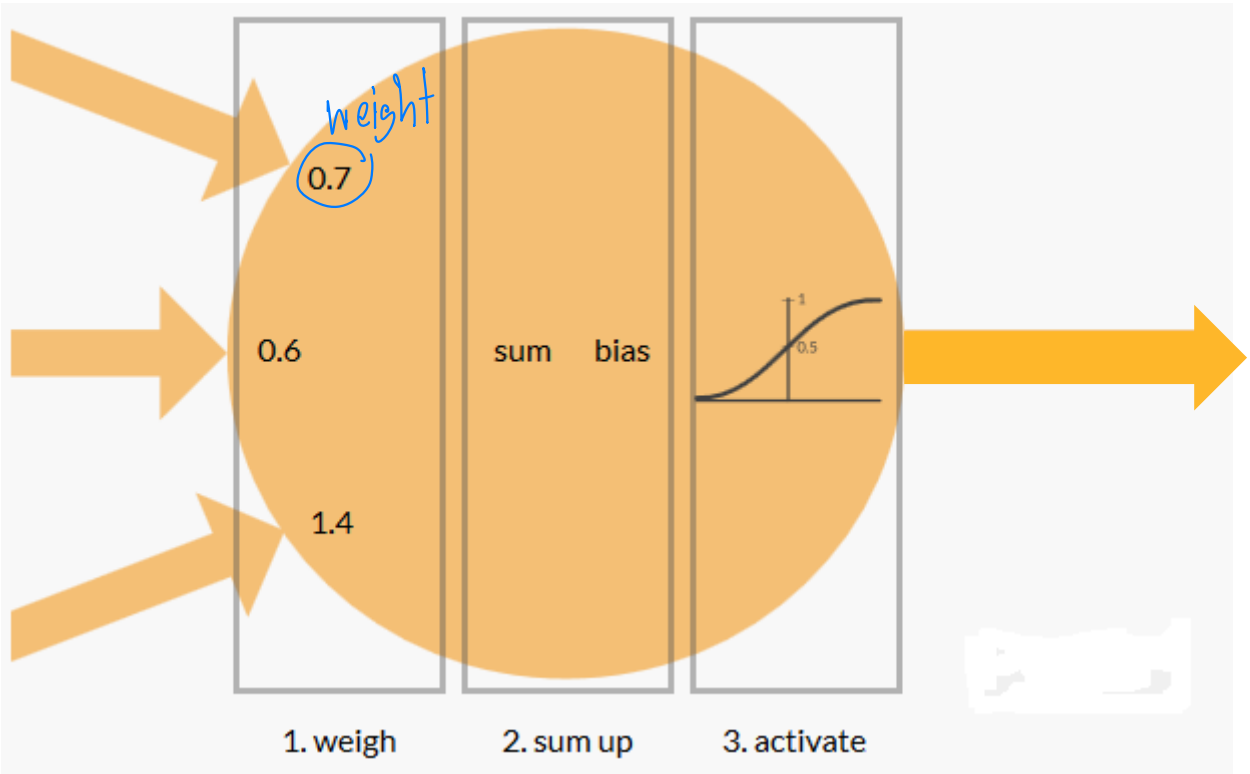
Artificial neuron





Inside artificial neuron

Input signals

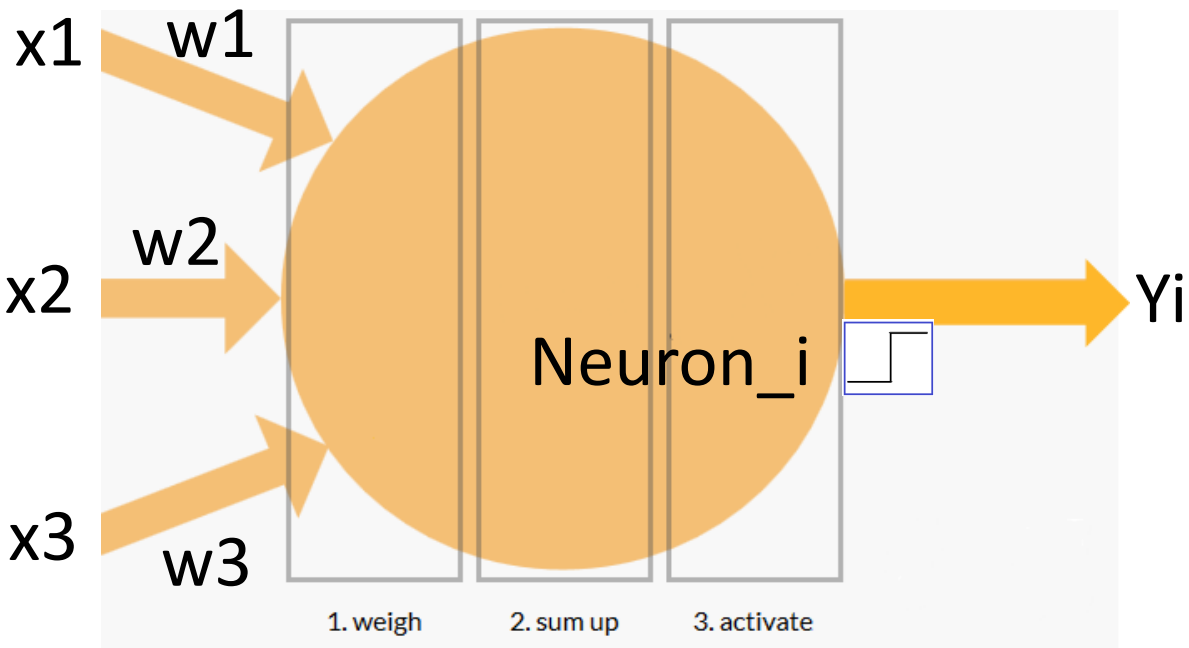


Output signal

We will call this artificial neuron as a **perceptron**.

Notations

Perceptron is the simplest form of a neural network. It consists of a single neuron with adjustable weights and a hard limit activation function.

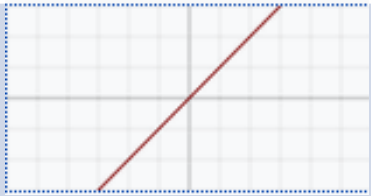
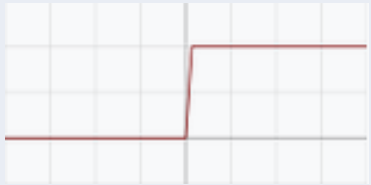
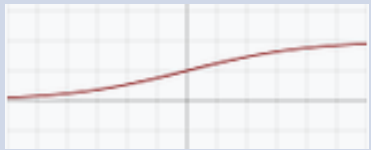
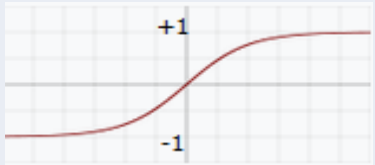


There are many kinds of activation function.

Frank Rosenblatt

invented perceptron in 1957

Activation functions

Name	Equation	Plot
Identity (Linear)	$f(x) = x$	
Binary step	$f(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ 1 & \text{for} \quad x \geq 0 \end{cases}$	
Sigmoid (Logistic)	$f(x) = \dfrac{1}{1 + e^{-x}}$	
TanH	$f(x) = \tanh(x) = \dfrac{2}{1 + e^{-2x}} - 1$	
Softmax	$f_i(\vec{x}) = \dfrac{e^{x_i}}{\sum_{j=1}^{J} e^{x_j}}$ for i = 1, ..., J	

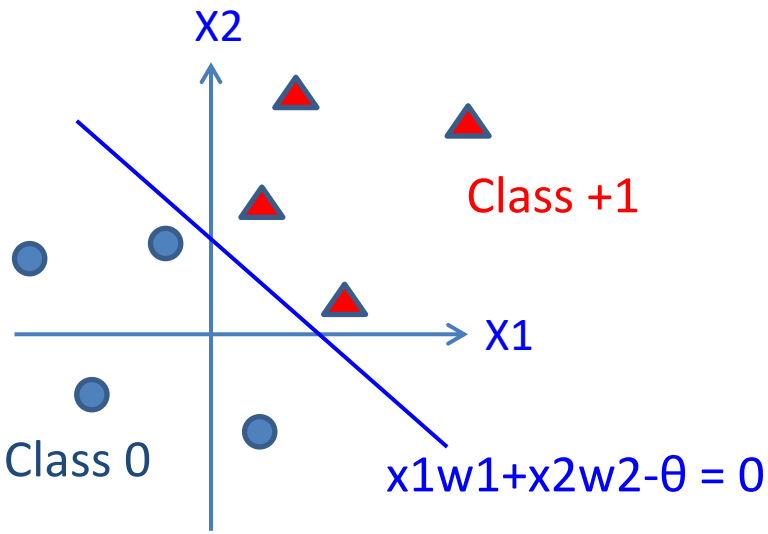
จุดต่างสุด

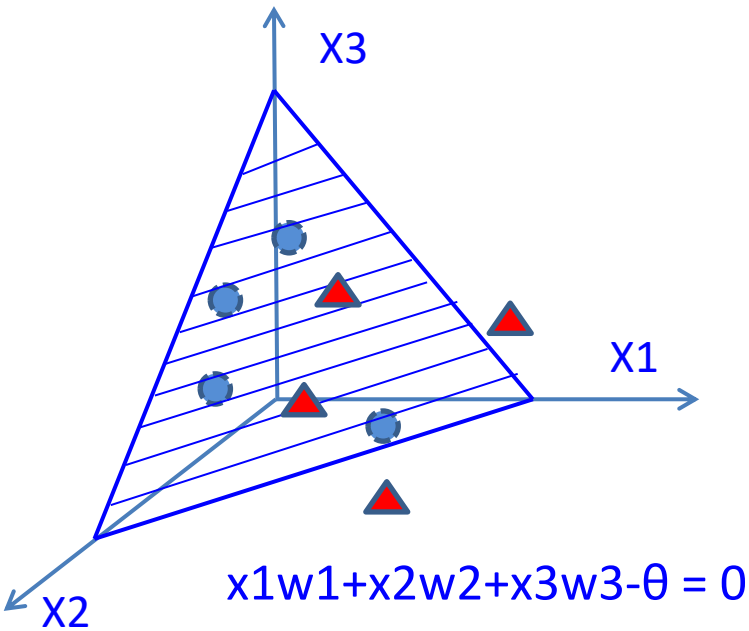
เกิด convergence ได้
เร็วกว่า sigmoid ถ้า loss ↑

Linear separability

แบ่งข้อมูล 2 class ให้ออกจากกันได้

Data in the n-dimensional space is **linearly separable** if two classes of data are divided by a hyperplane into two decision regions.





In general, the hyperplane is defined by the linearly separable function.

$$\sum_{i=1}^{n} x_i w_i - \theta = 0$$

(The threshold θ can be used to shift the decision boundary.)

- The perceptron learns its classification task by making small adjustments in the weights to reduce the difference between the actual and desired (target) outputs of the perceptron.

Target value Actual value

$$e(p) = Y_d(p) - Y(p)$$

Where

p is the training pattern (example)

$Y_d(p)$ is the desired (target) output of pattern p

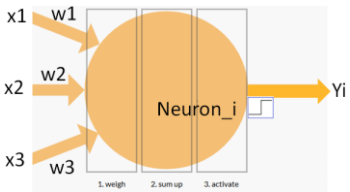
Y(p) is the actual output

e(p) is the difference (error) between $Y_d(p)$ and Y(p)

- It uses e(p) to update weights of the next iteration,

$$w_i(p+1) = w_i(p) - \alpha . x_i(p) . e(p)$$

where α is the learning rate (0~1)



Perceptron learning algorithm

Step 1: Initialization

- Set initial weights w1,w2,…,wn and thresholds (θ) to small random numbers in the range [-0.5,+0.5]
- p = 1 # the first pattern

Step 2: Activation

$$Y(p) = step\left(\sum_{i=1}^{n} x_i(p).w_i(p) - \theta\right)$$

where n is the number of perceptron inputs.

Step 3: Weight training by using gradient descent

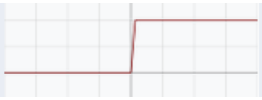
$$w_i(p+1) = w_i(p) + \Delta w_i(p),$$

$$\Delta w_i(p) = -\alpha.x_i(p).e(p)$$ where $e(p) = Yd(p) - Y(p)$

Step 4: Iteration

Increase p by one, go back to step 2 until each pattern is trained.

Step 5: If the perceptron doesn't converge, p = 1 and repeat steps 2 – 4.

Binary step	$f(x) = \begin{cases} 0 & \text{for} \quad x < 0 \\ 1 & \text{for} \quad x \geq 0 \end{cases}$	

$W_i(p+1) = 0.3 - (0.1 \cdot 1) \cdot -1$

$= 0.3 - (0.1 \cdot 1 \cdot (-1))$

$= 0.3 + 0.1$

$= 0.4$

$yp = ((0.3 \cdot 1) + (0 \cdot -0.1)) - \theta$

$= 0.3 - 0.3$

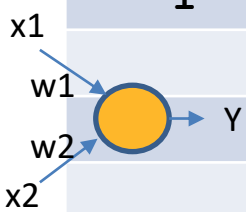
$= 0$

Af activation Binary step $= 1$ ✓

Example: Train a perceptron on AND

$$w_i(p+1) = w_i(p) - \alpha . x_i(p) . e(p)$$

θ= 0.3, α = 0.1



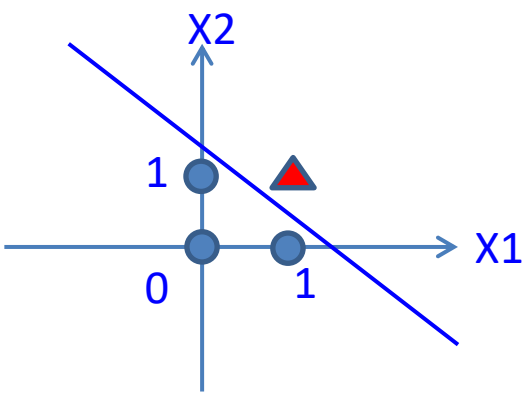
Epoch	Inputs x1	Inputs x2	Yd (desired)	Weights w1	Weights w2	Y (actual)	Error	Weights w1	Weights w2
1	0	0	0	0.3	-0.1	0	0	0.3	-0.1
	0	1	0	0.3	-0.1	0	0	0.3	-0.1
	1	0	0	0.3	-0.1	1	-1	0.4	-0.1
	1	1	1	0.4	-0.1	1	0	0.4	-0.1
2	0	0	0	0.4	-0.1	0	0	0.4	-0.1
	0	1	0	0.4	-0.1	0	0	0.4	-0.1
	1	0	0	0.4	-0.1	1	-1	0.3	-0.1
	1	1	1	0.3	-0.1	0	1	0.2	-0.2
.	.		.	.		.	.	.	
.	.		.	.		.	.	.	
.	.		.	.		.	.	.	
5	0	0	0	0.15	0.15	0	0	0.15	0.15
	0	1	0	0.15	0.15	0	0	0.15	0.15
	1	0	0	0.15	0.15	0	0	0.15	0.15
	1	1	1	0.15	0.15	1	0	0.15	0.15

1 epoch

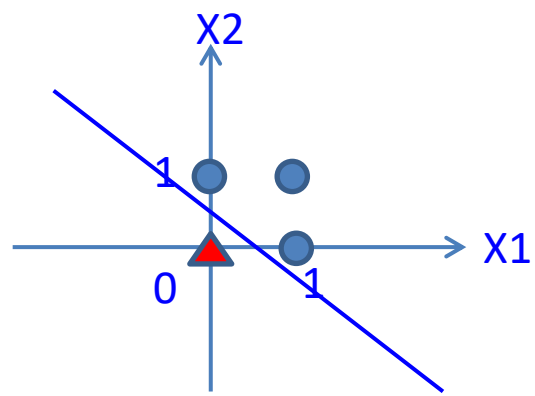
แสดง weight ที่ได้ train ไม่เสร็จ

Problem of perceptron

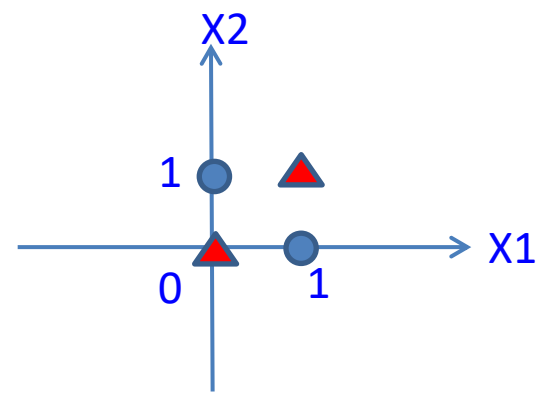
- Perceptron can learn only simple linear separable problems, e.g., AND, OR. It failed on XOR problem.



X1 AND X2



X1 OR X2



X1 XOR X2

- A single perceptron can classify only linear separable problems, regardless of whether we use a hard-limit or soft-limit activation functions.
- Moreover, increasing the number of perceptrons in the same layer doesn't help.

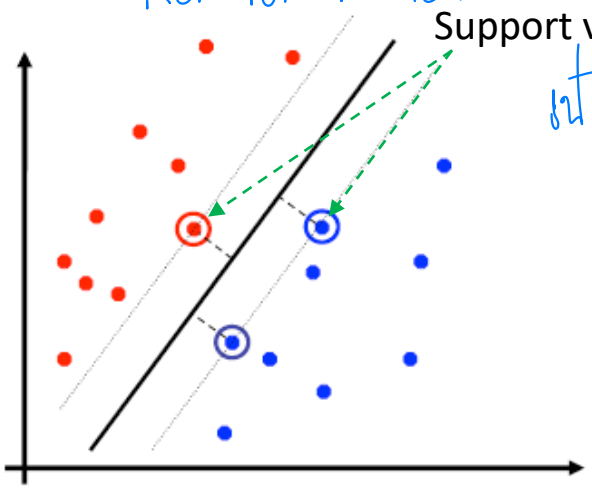
Linear Classification by Support Vector Machine

- SVMs (Vapnik, 1990's) choose the linear separator with the **largest margin**

SVM มี kernel function มีหน้าที่แปลง ปัญหาที่เป็น non linear ให้เป็น linear

Support vectors

Robust to outliers!



V. Vapnik

- Good according to intuition, theory, practice
- SVM became famous when, using images as input, it gave accuracy comparable to neural-network

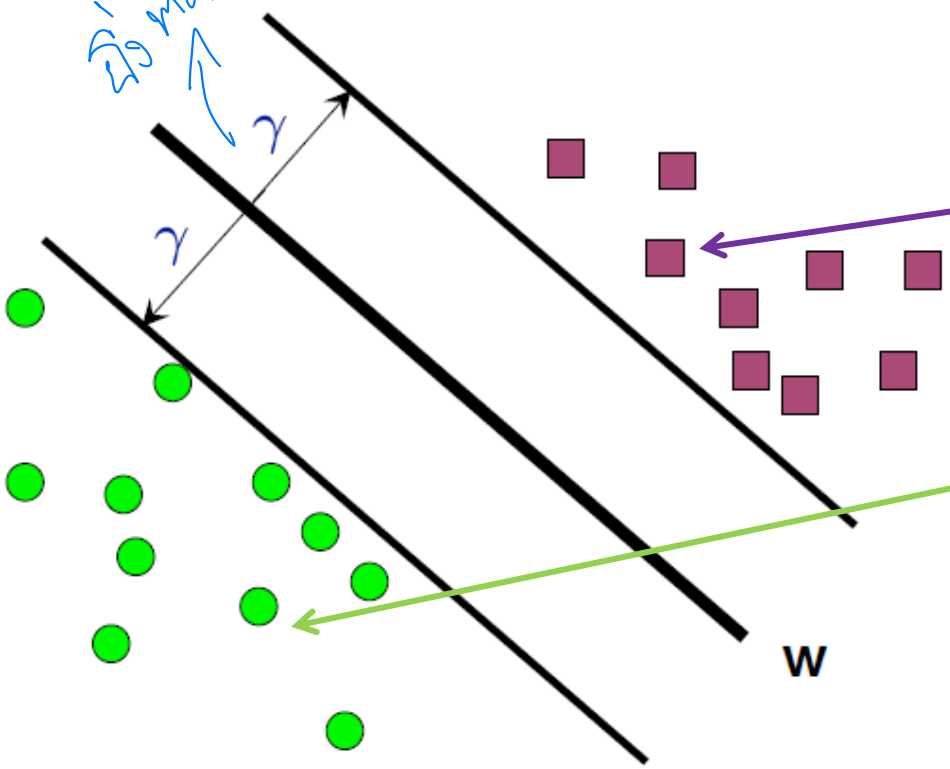
Pages 22-24 are taken from Introduction To Machine Learning @MIT

Linear Classification by Support Vector Machine

$\exists \mathbf{w}$ such that $\forall t$ $\quad y_t(\mathbf{w} \cdot \mathbf{x}_t) \geq \gamma > 0$

Called the *margin*

ยิ่ง margin เยอะยิ่งดี



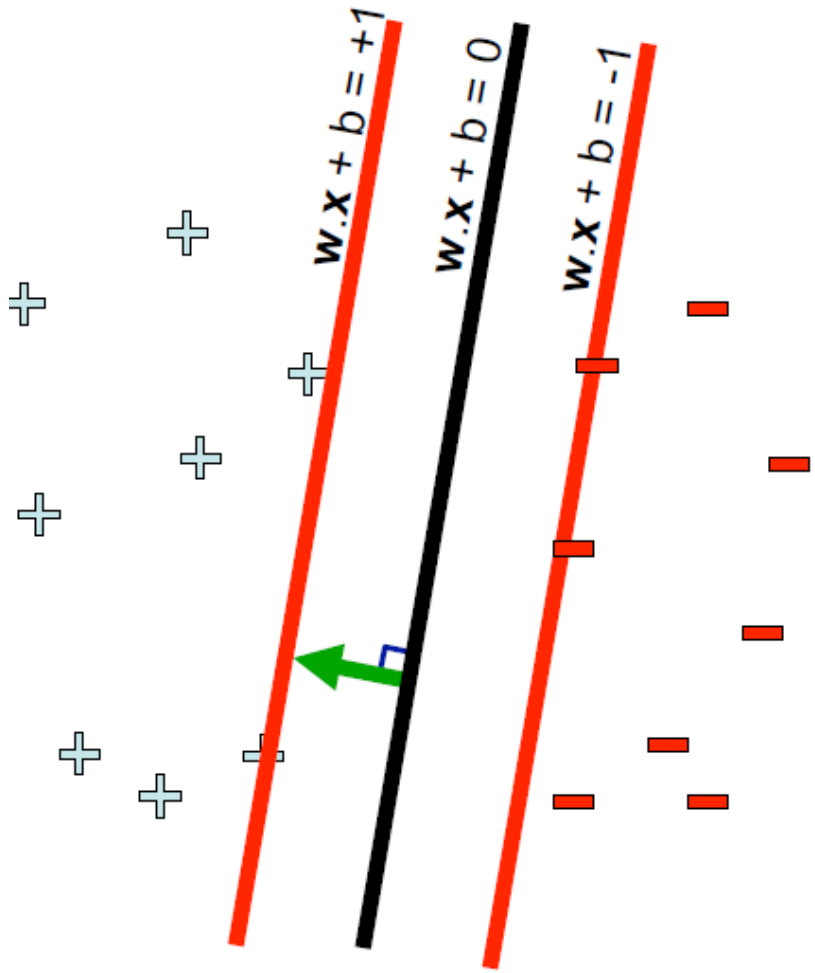
Equivalently, for $y_t = +1$,

$$w \cdot x_t \geq \gamma$$

and for $y_t = -1$,

$$w \cdot x_t \leq -\gamma$$

Linear Classification by Support Vector Machine



Suppose that the value of margin is 1.

We are asked to find a set of weights (w) such that, for all pattern t,

for $y_t = +1$, $w \cdot x_t + b \geq 1$

and for $y_t = -1$, $w \cdot x_t + b \leq -1$

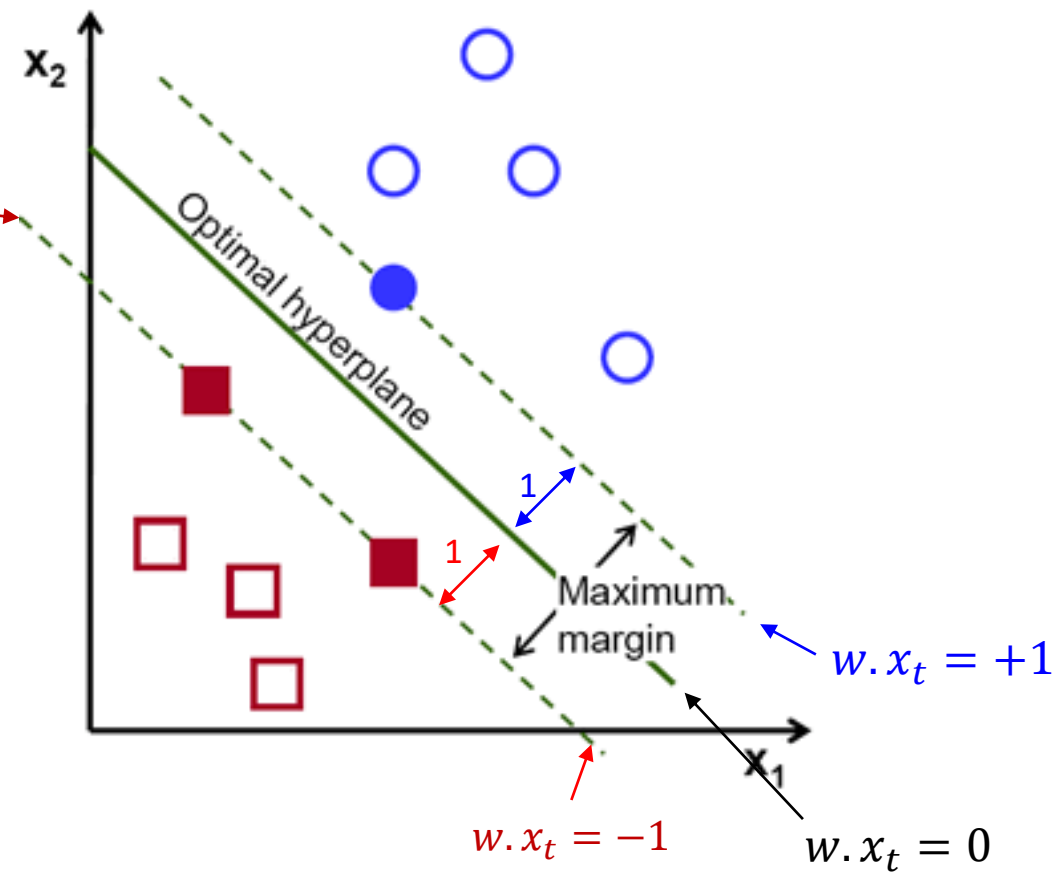
That is, we want to satisfy all of the **linear** constraints

$$y_t \left(w \cdot x_t + b \right) \geq 1 \quad \forall t$$

SVM Loss

- Loss of SVM

Any rectangle instance that crosses the maximum margin (dash line) will be penalized.

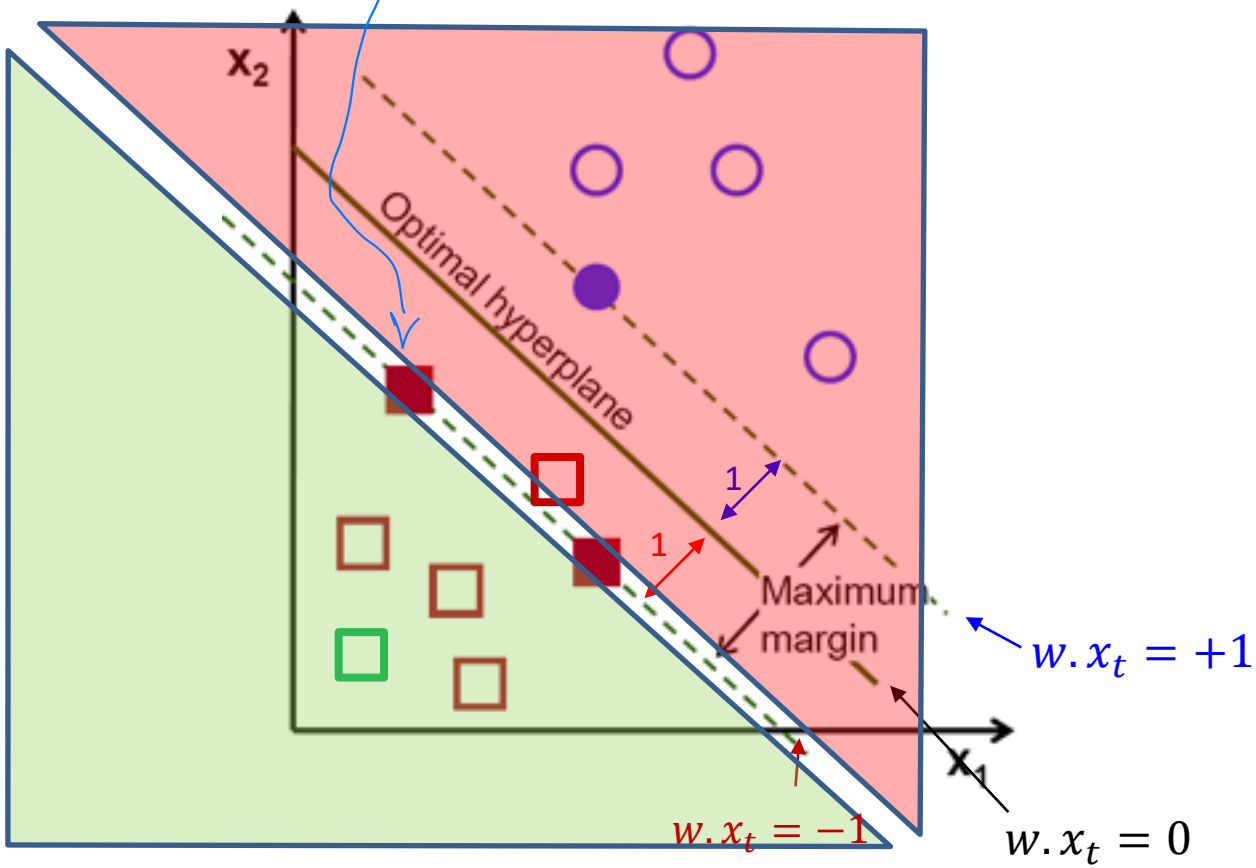


SVM Loss

ถ้าอยู่ไม่ไกล optimal hyperplane
เท่ากับ ยังไม่เกิด loss

Penalty = 0

Penalty > 0



SVM Loss

Let $s = f(x_i, w)$; score of input x_i

s_{yi} is the score of the target class

Score of other class

the SVM loss has the form:

Score of the target class

margin

loss

$$L_i = \sum_{j \neq y_i} \begin{cases} 0 & \text{if } s_{y_i} \geq s_j + 1 \\ s_j - s_{y_i} + 1 & \text{otherwise} \end{cases}$$

$$= \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)$$

	S_{yi}
cat	3.2
car	S_j 5.1
frog	S_j -1.7

$$\boxed{L_i = \sum_{j \neq y_i} \max(0, s_j - s_{y_i} + 1)}$$

= max(0, 5.1 - 3.2 + 1)
+max(0, -1.7 - 3.2 + 1)
= max(0, 2.9) + max(0, -3.9)
= 2.9 + 0
= 2.9

- Next class
  - Multilayer neural networks